



Coursework Brief and Front Sheet PGT

This front sheet for assessments is designed to contain the brief, the submission instructions, and the actual student submission for any WMG assessment. As a result, the sheet is completed by several people over time and is therefore split up into sections explaining who completes what information and when. Yellow highlighted text indicates examples or further explanation of what is requested, and the highlight and instructions should be removed as you populate 'your' section. This sheet is only to be used for components of assessment worth more than 3 CATS (e.g. for a 15 credit module, weighted more than 20%; or for a 10 credit module, weighted more than 30%).

To be completed (highlighted parts only) by the Programme Administration after approval and prior to issuing of the assessment; to be consulted by the Student(s) so that you know how and when to submit:

Date set	May 2026
Submission date (excluding extensions)	Tuesday 15th September 2026.
Submission guidance	The report and notebook must be uploaded to Tabula.

To be completed by the Module Leader/Tutor prior to approval and issuing of the assessment; to be briefed to the Student(s) so that they understand the Coursework Brief, its context within the module, and any specific criteria and advice from the tutor:

Module Title & Code	WMG9B7-15 - Artificial Intelligence & Deep Learning
Component Type	Project
Module Leader	Dr Leonardo Alves Dias
Module Tutor	See above
Assessment Title	Individual Reassignment
Weighting (%)	70%



Coursework Brief

Don't use the content from this assessment in your dissertation

Artificial Intelligence (AI) and Deep Learning (DL) are reshaping how problems are solved across every sector, including transportation and mobility. As AI capabilities continue to evolve, professionals must not only be able to apply algorithms effectively but also understand the technical rationale, implications, and emerging trends shaping the future of the field.

In this assessment, you will take on the role of an AI practitioner tasked with developing a small-scale DL application to solve a real-world problem using open-source data. Besides, you will submit a reflective report that critically analyses your design choices, compares your approach to traditional machine learning methods, and explores the broader implications and sustainability of your work.

Therefore, you must do the following two parts for this assessment:

Part 1: AI or Deep Learning Application

Develop a functional DL solution using Python that addresses a well-defined problem in transportation and mobility using a suitable open-source dataset (subject to approval). You must identify and select a clearly defined problem, and your solution must address it and demonstrate its real-world relevance.

- As a result, you must submit a reproducible Jupyter Notebook with clear documentation and comments that contains appropriate methods, e.g., data preprocessing, model selection, training, and evaluation, among others. Also, your notebook must contain a README cell to guide markers on how to run your code.

You may incorporate pre-trained architectures as appropriate, but you must demonstrate thoughtful design and configuration of your solution, reflecting your understanding of the tools and algorithms used. Although AI assistance is allowed during development, you must adapt and extend any AI-generated content to reflect your understanding and ownership of the work.

Note: Your solution must be related to transportation and mobility, e.g., urban traffic analysis.

Part 2: Critical Reflection

You will write a structured critical reflection that analyses and contextualises your implementation from Part 1. Your reflection must address the following areas:

- Explain and compare your chosen solution (for your chosen problem) with a traditional machine learning technique and justify the use of deep learning or AI in your project.
- Reflect on the key design decisions you made, including (but not limited to) model architecture, parameters, and evaluation strategy. Besides, discuss how you used AI assistants (if applicable) to support your implementation and how you expanded or adapted the generated output.
- Discuss the potential impact of your solution on individuals, organisations, environment or society based on the relevance to your chosen problem.
- Discuss future trends and opportunities that can apply to your problem and solution. For example, you can identify one or more emerging trends in AI or deep learning and explore how such trends could enhance or transform your solution if implemented in future iterations.

References from both academic and commercial sources should be included to support your discussion. Ensure your report is clear, objective, and well-structured, adhering to academic standards.



Additional document will be accessible from the module's Moodle site, as well as an FAQ.

You can use AI tools in your workflow. However, your application must clearly reflect your own reasoning and decision-making, critical evaluation, original coding effort, design and implementation decisions. Although AI assistance is allowed during code development, you must adapt and extend any AI-generated content to reflect your understanding and ownership of the work. This aims to bridge applied AI development with responsible practice, ethical reasoning, and awareness of industry and research trends.

You cannot use the same project submitted as your original assessment.

Assessment Length/ Word count	The maximum word count is 2800 words. The word count figure includes quotations, tables, figures, footnotes, endnotes, in-text citations, titles, abstracts & summaries. It does not include the questions as set, or other material from the rubric or supplementary information such as tables of contents, tables of figures, tables of tables, the reference list, or bibliographies. Material submitted in an annex or appendix is also outside the word count. There is +10% allowed on the word count without penalty. If you exceed the word count by more than 10% to 30%, a penalty of 10% on the original mark awarded will be applied from the original total mark but capped at the pass mark. If you exceed the word count by more than 30%, the final mark will be capped at the pass mark.		
ARTIFICIAL INTELLIGENCE (AI)			
AI Scale (Please copy and paste this tick '✓' into the appropriate box)		NO AI	You must not use AI at any point during the assessment. You must demonstrate your core skills and knowledge.
		AI PLANNING	You may use AI for planning, idea development, and research. Your final submission should show how you have developed and refined these ideas.
	✓	AI COLLABORATION	You may use AI to assist with specific tasks such as drafting text, refining and evaluating your work. You must critically evaluate and modify any AI-generated content you use.
		FULL AI	You may use AI extensively throughout your work either as you wish, or as specifically directed in your assessment. Focus on directing AI to achieve your goals while demonstrating your critical thinking.



		AI EXPLORATION	You should use AI creatively to solve the task.
AI Policy	Unauthorised and undisclosed use of Artificial Intelligence may result in a finding of Academic Misconduct and an academic sanction under Regulation 11 . Guidance for WMG student on the use of Artificial Intelligence can be found here .		
Module Learning Outcomes (numbered)	LO1. Critically evaluate and explain the key differences between traditional machine learning and deep learning approaches. LO2. Critically evaluate deep learning and artificial intelligence solutions and their implications. LO3. Demonstrate an understanding of emerging trends and research development in AI. LO4. Design and evaluate deep learning models for practical applications. LO5. Collaborate effectively as part of a group to develop and communicate a practical AI solution by presenting findings through reproducible code, visualisations, or technical reporting.		
Learning Outcomes assessed in this assessment (numbered)	LO1, LO2, and LO3.		
Learning Outcomes	Please remember that you must meet all Learning Outcomes to achieve a pass mark.		
Late Submission Policy	If work is submitted late, penalties will be applied at the rate of 5 marks per University working day after the due date, up to a maximum of 10 working days late. After this period, the mark for the work will be reduced to 0 (which is the maximum penalty). “Late” means after the submission deadline time as well as the date – work submitted after the given time even on the same day is counted as 1 day late.		
Resit Policy	If you fail this module and/or component, the University allows students to remedy failure (within certain limits). Decisions to authorise resits are made by Exam Boards. These will be issued at specific times of the year, depending on your programme of study. More information can be found from your programme office if you are concerned. If this is already a resit attempt, this means you will not be eligible for an additional attempt. The University allows as standard a maximum of two attempts on any assessment (i.e. only one resit). Students can only have a third attempt under exceptional circumstances via a Mitigating Circumstances Panel decision.		
Retention of Drafts & Records	Please ensure that you retain any drafts of your work, associated notes, records of research should they be required by the marker or moderator. You do not need to submit these, but they should be retained until after the end of your registration.		

**Where to get help:**

1. Talk to your module tutor if you do not understand the question or are unsure as to exactly what is required.
2. There are also numerous online courses provided by the University library to help in academic referencing, writing, avoiding plagiarism and a number of other useful resources.
<https://warwick.ac.uk/services/library/students/your-library-online/>
3. If you have a problem with your wellbeing, it is important that you contact your personal tutor or wellbeing support services <https://warwick.ac.uk/services/wss>



Marking Guidelines:

Learning Outcome or Evaluative Criteria	80+ Outstanding	70-79 Distinction	60-69 Good Pass	50-59 Pass	40-49 Marginal Fail	<40 Fail
LO1. Critically evaluate and explain the key differences between traditional machine learning and deep learning approaches.	An exemplary answer, showing complete mastery of the learning outcome, with an exceptionally developed and mature ability to analyse, synthesise and apply concepts, models, and techniques. All requirements of the learning outcome are exceeded, and the answer is free from errors. The answer demonstrates originality of thought, with strong critical reflection and the ability to tackle issues not previously encountered. Ideas are explained with great lucidity and in an extremely structured manner. Sophisticated and in-depth comparison of ML and DL paradigms.	An excellent answer, showing mastery of the learning outcome, with a highly developed and mature ability to analyse, synthesise and apply concepts, models, and techniques. All requirements of the learning outcome are covered, and work is free from all but very minor errors. There is good critical reflection and the ability to tackle issues not previously encountered. Ideas are explained very clearly and in a highly structured manner. Strong, critical comparison that reflects sound theoretical and practical knowledge. Justification for the chosen DL approach is well-integrated with the problem context.	A strong answer, showing a sound grasp of the learning outcome and a skillful attempt at analysis, synthesis and application of concepts, models, and techniques. All requirements of the learning outcome are covered, but there may be some minor errors. There is some critical reflection, and a reasonable attempt is made to tackle issues not previously encountered. Ideas are explained clearly and in a well organised manner, with some minor exceptions. Offers a clear, accurate, and well-reasoned comparison. Demonstrates good technical understanding and articulates relevant trade-offs linked to the problem.	A satisfactory answer, showing a grasp of the learning outcome but analysis, synthesis and application of concepts, models and techniques is mechanical, with a heavy reliance on course materials. The requirements of the learning outcome are covered but with little or no critical reflection and limited ability to tackle issues not previously encountered. Ideas are explained adequately but with some confusion and lack of organisation. Provides a basic but valid explanation of ML vs. DL, with some relevant distinctions and generic justification for choosing DL.	An unsatisfactory answer. The learning outcome is not met, but it would not take too much to meet it. There is a weak attempt at analysis, synthesis and application of concepts, models, and techniques. Only some of the requirements of the learning outcome are covered. Inability to reflect critically and difficulty in beginning to address issues not previously encountered. Ideas are poorly explained and organised. Shows limited or partial understanding. Comparison is superficial, with some factual inaccuracies or weak justification for the chosen method.	An inadequate answer and there are extremely serious gaps in knowledge and many areas of confusion. Few or none of the requirements of the learning outcome are covered. There is a lack of serious engagement with the learning outcome and there is no evidence that issues not previously encountered can be addressed. The levels of expression and organisation in the work are very inadequate. Demonstrates little or no understanding of the differences between ML and DL. No meaningful comparison. Terminology and conceptual foundations are incorrect or missing.

	Demonstrates advanced understanding, uses precise technical language, and presents a highly contextualised justification for the approach taken well-integrated with the problem context.				Lack or insufficient references and citations.	Lack or insufficient references and citations.
LO2. Critically evaluate deep learning and artificial intelligence solutions and their implications.	An exemplary answer, showing complete mastery of the learning outcome, with an exceptionally developed and mature ability to analyse, synthesise and apply concepts, models, and techniques. All requirements of the learning outcome are exceeded, and the answer is free from errors. The answer demonstrates originality of thought, with strong critical reflection and the ability to tackle issues not previously encountered. Ideas are explained with great lucidity and in an extremely structured manner.	An excellent answer, showing mastery of the learning outcome, with a highly developed and mature ability to analyse, synthesise and apply concepts, models, and techniques. All requirements of the learning outcome are covered, and work is free from all but very minor errors. There is good critical reflection and the ability to tackle issues not previously encountered. Ideas are explained very clearly and in a highly structured manner. Strong implementation with thoughtful design and use of tools, and well documented. Evaluation is	A strong answer, showing a sound grasp of the learning outcome and a skillful attempt at analysis, synthesis and application of concepts, models, and techniques. All requirements of the learning outcome are covered, but there may be some minor errors. There is some critical reflection, and a reasonable attempt is made to tackle issues not previously encountered. Ideas are explained clearly and in a well organised manner, with some minor exceptions. Solution is well-structured, reproducible, and reasonably	A satisfactory answer, showing a grasp of the learning outcome but analysis, synthesis and application of concepts, models and techniques is mechanical, with a heavy reliance on course materials. The requirements of the learning outcome are covered but with little or no critical reflection and limited ability to tackle issues not previously encountered. Ideas are explained adequately but with some confusion and lack of organisation. Implementation is functional and includes the main pipeline stages. Code is reproducible, though may lack clarity or	An unsatisfactory answer. The learning outcome is not met, but it would not take too much to meet it. There is a weak attempt at analysis, synthesis and application of concepts, models, and techniques. Only some of the requirements of the learning outcome are covered. Inability to reflect critically and difficulty in beginning to address issues not previously encountered. Ideas are poorly explained and organised. The code runs but is poorly structured or barely functional. Key	An inadequate answer and there are extremely serious gaps in knowledge and many areas of confusion. Few or none of the requirements of the learning outcome are covered. There is a lack of serious engagement with the learning outcome and there is no evidence that issues not previously encountered can be addressed. The levels of expression and organisation in the work are very inadequate. Implementation is missing, incomplete, or fails to address the stated problem. Code lacks structure or

	Technically exemplary and professional-level implementation alongside documentation. Evaluation is thorough, insightful, and well-justified. Ethical, ecological, and societal implications are critically analysed with maturity, originality, and contextual depth.	comprehensive and includes interpretability. The reflection presents a clear, critical discussion of implications.	documented. Evaluation of the model is clear and includes interpretability. The written reflection explores ethical, societal, and/or environmental considerations with insight.	detail. Evaluation uses some appropriate metrics. Ethical or contextual implications are acknowledged, though analysis is basic.	stages (e.g. preprocessing, training, evaluation) are underdeveloped. Evaluation of solution is vague. Ethical and social implications are only briefly mentioned. Lack or insufficient references and citations.	documentation. Evaluation is absent or inappropriate. Ethical and contextual considerations are missing or confused. Lack or insufficient references and citations.
LO3. Demonstrate an understanding of emerging trends and research development in AI.	An exemplary answer, showing complete mastery of the learning outcome, with an exceptionally developed and mature ability to analyse, synthesise and apply concepts, models, and techniques. All requirements of the learning outcome are exceeded, and the answer is free from errors. The answer demonstrates originality of thought, with strong critical reflection and the ability to tackle issues not previously encountered.	An excellent answer, showing mastery of the learning outcome, with a highly developed and mature ability to analyse, synthesise and apply concepts, models, and techniques. All requirements of the learning outcome are covered, and work is free from all but very minor errors. There is good critical reflection and the ability to tackle issues not previously encountered. Ideas are explained very clearly and in a highly structured manner.	A strong answer, showing a sound grasp of the learning outcome and a skillful attempt at analysis, synthesis and application of concepts, models, and techniques. All requirements of the learning outcome are covered, but there may be some minor errors. There is some critical reflection, and a reasonable attempt is made to tackle issues not previously encountered. Ideas are explained clearly and in a well organised manner, with some minor exceptions.	A satisfactory answer, showing a grasp of the learning outcome but analysis, synthesis and application of concepts, models and techniques is mechanical, with a heavy reliance on course materials. The requirements of the learning outcome are covered but with little or no critical reflection and limited ability to tackle issues not previously encountered. Ideas are explained adequately but with some confusion and lack of organisation. Identifies at least one relevant trend or	An unsatisfactory answer. The learning outcome is not met, but it would not take too much to meet it. There is a weak attempt at analysis, synthesis and application of concepts, models, and techniques. Only some of the requirements of the learning outcome are covered. Inability to reflect critically and difficulty in beginning to address issues not previously encountered. Ideas are poorly explained and organised.	An inadequate answer and there are extremely serious gaps in knowledge and many areas of confusion. Few or none of the requirements of the learning outcome are covered. There is a lack of serious engagement with the learning outcome and there is no evidence that issues not previously encountered can be addressed. The levels of expression and organisation in the work are very inadequate.



	Ideas are explained with great lucidity and in an extremely structured manner. Demonstrates outstanding understanding of cutting-edge AI trends and research. Shows originality in proposing how trends could significantly evolve or transform the current solution or field.	Strong awareness of current AI research and innovation. Trends are contextualised, with clear links to the problem domain and project future.	Good understanding of relevant AI developments. Clearly explains one or more trends and how they could shape or enhance future work.	research direction. Some attempt to relate it to the problem and solution.	Some awareness of AI trends, but discussion lacks clarity or accuracy. Connection to the chosen problem is weak or unclear. Lack or insufficient references and citations.	No meaningful reference to current AI trends or research. Mentions are vague or incorrect. No link to the project context. Lack or insufficient references and citations.
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